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cIPMA: An R package for combined importance-performance map analysis

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ABSTRACT

Combined importance–performance map analysis (cIPMA) extends partial least squares structural equation modeling (PLS-SEM) by integrating importance as sufficiency estimates with bottleneck diagnostics necessity thresholds from Necessary Condition Analysis (NCA). While conceptually powerful, the current workflow for cIPMA is fragmented, typically requiring commercial software, manual data rescaling, and external spreadsheet transformation and visualization, which increase the risk of human error and reduce reproducibility. This article introduces `cIPMA`, an open-source R package that automates the complete cIPMA workflow on top of `seminr` for PLS-SEM and `NCA` for necessary condition analysis. The package implements the rescaling and weighting procedures required for importance–performance maps, calls NCA's CE-FDH ceiling technique to obtain necessity effect sizes and bottleneck tables, and provides publication-ready visualization. Using the `cIPMA` package, researchers can perform a full cIPMA analysis in a single reproducible script and reduce the likelihood of specification and transcription errors. The package provides a transparent, open-source implementation that enables researchers to explore the interplay between probabilistic sufficiency and necessity logics in behavioral research while adhering to the established methodological requirements of cIPMA.

Metadata

Code metadata.

Nr.	Code metadata description	Metadata
C1	Current code version	v0.1.6
C2	Permanent link to code/repository used for this code version	https://github.com/MarekDejaUJ/cIPMA
C3	Permanent link to Reproducible Capsule	
C4	Legal Code License	GNU General Public License (GPL) v3
C5	Code versioning system used	Git
C6	Software code languages, tools, and services used	R
C7	Compilation requirements, operating environments & dependencies	R (>= 4.0.0), Dependencies: seminr, NCA, ggplot2, ggrepel, stats
C8	If available Link to developer documentation/manual	https://github.com/MarekDejaUJ/cIPMA/blob/main/README.md
C9	Support email for questions	marek.deja@uj.edu.pl

1. Motivation and significance

In management and social science research, analysts often want to prioritize which predictors of a focal construct (e.g., technology acceptance or customer satisfaction) should be targeted by interventions. Importance–Performance Map Analysis (IPMA) within Partial Least Squares Structural Equation Modeling (PLS-SEM) is a popular tool for this purpose [1]. In IPMA, latent composite scores are interpreted as performance indicators, while total effects from the structural model express "importance" in a probabilistic sufficiency sense: predictors are attractive levers when higher scores are associated, on average, with higher levels of the outcome [2].

This sufficiency logic, however, does not capture constraints that must be satisfied before high outcomes become feasible. Necessary Condition Analysis (NCA) addresses such constraints by identifying "must-have" conditions: if a necessary antecedent does not reach a minimum level, high outcomes are structurally impossible, regardless of

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other variables [3]. Combined Importance–Performance Map Analysis (cIPMA) [1] links these perspectives by overlaying NCA results on an IPMA map. Necessary predictors are highlighted, and the size of their symbols reflects the proportion of cases that fail to reach the required level for a chosen target outcome. This joint display allows researchers and practitioners to separate predictors that should be improved because they have large average effects from predictors that must be improved because they form bottlenecks [1].

In practice, the current cIPMA workflow remains fragmented. The eight-step procedure proposed by Hauff et al. [1] is implemented in SmartPLS 4 and typically requires estimating a PLS-SEM model, exporting rescaled latent scores, running NCA with the CE-FDH ceiling technique, and manually assembling effect sizes and bottleneck information in a spreadsheet. These GUI-based exports, re-imports, and manual transformations increase the risk of avoidable errors (e.g., inconsistent rescaling or misaligned variables) and complicate reproducibility and automation. Moreover, standard PLS-SEM assumes that the structural model is correctly specified by the researcher; if it is misspecified, the "importance" implied by total effects is not causally interpretable [4,5]. NCA, by contrast, isolates bivariate necessity relations. When NCA and PLS-SEM are combined without careful model justification, redundant or spurious paths may be interpreted as important or necessary, creating an avoidable threat to inference.

Accordingly, cIPMA—like any analytical method—rests on methodological requirements that must be met for results to be interpretable. At minimum, researchers should ensure that (1) the PLS-SEM model is theoretically justified and properly specified, (2) outer weights are positive to support meaningful performance rescaling, (3) indicators have well-defined theoretical minima and maxima for the 0–100 transformation, and (4) the sample size is adequate for both PLS-SEM estimation and NCA permutation testing [1].

In addition, Dul's intervention-oriented framing highlights that SEM-based recommendations are typically interpreted under a *ceteris paribus* assumption—changing one antecedent while holding others fixed—making IPMA and cIPMA essentially one-predictor-at-a-time tools for prioritization [6]. NCA's logic, in turn, is boundary-based and necessity-oriented rather than average sufficiency [2]. Researchers should therefore avoid conflating importance (sufficiency) with necessity (constraints), and interpret cIPMA results in light of both logics and the validity of the underlying theoretical model [2,6].

Despite growing interest in combining sufficiency and necessity perspectives, no R package automates the full cIPMA procedure in an open-source, scriptable environment. While NCA is available via the `NCA` package and composite-based PLS-SEM can be estimated with `seminr` [7], reproducing cIPMA still requires bespoke glue code, careful manual rescaling, and *ad hoc* integration of PLS-SEM and NCA outputs.

The `cIPMA` package addresses these gaps by re-implementing the SmartPLS-based workflow on top of `seminr` and `NCA`, so that the entire procedure can be executed from a single R script [1]. It reduces opportunities for human error by automating rescaling, necessity testing, and bottleneck computation, and it generates publication-ready plots directly from the analysis object. To support methodologically sound use, the package also implements `check_assumptions()`, which provides clear diagnostics for key prerequisites (positive outer weights, reliability thresholds, discriminant validity via HTMT, multicollinearity via VIF, and sample size adequacy). Finally, motivated by recent work showing that discovery-oriented methods can complement theory-driven PLS-SEM in exploratory contexts [8], the package includes an explicitly experimental `discovery_cipma()` module based on `bnlearn` [9,10] for exploratory model comparison. Because no equivalent methodological framework currently formalizes how probabilistic structure learning should be integrated with NCA's boundary-based necessity logic, this component is intentionally framed as future research rather than a validated part of the core cIPMA workflow.

2. Software description

The `cIPMA` package is intended as a compact toolset for performing cIPMA within the R ecosystem. It targets researchers who already use PLS-SEM and NCA, and who want to move from menu-driven, multi-software workflows to a reproducible script that covers estimation, necessity analysis, and visualization. Conceptually, the package links two components: PLS-SEM (via `seminr`) and NCA (via `NCA`). Fig. 1 provides a schematic overview of the cIPMA workflow, illustrating how the package integrates the sufficiency path (PLS-SEM total effects) with the necessity path (NCA bottlenecks) to produce the final cIPMA output.

Users specify the measurement and structural models in `seminr`'s `lavaan`-like syntax (for example, the `lavaan` statement `Perceived_Usefulness =~ PU_01 + PU_02 + PU_03` corresponds to a composite block in `seminr`), supply the raw data matrix, and provide the theoretical minimum and maximum for each indicator. They also identify a target construct for cIPMA and choose a target outcome level on the 0–100 scale for the bottleneck analysis.

On this basis, `cIPMA` assumes that a PLS-SEM model has been estimated and bootstrapped with `seminr`. The core function `cipma()` takes the bootstrapped model, the raw data, and the list of indicator scales as arguments. It extracts total effects from the bootstrapped path model, which serve as the "importance" dimension, reconstructs latent variable scores using the estimated outer weights, rescales these scores to a common 0–100 "performance" scale, and then calls `NCA::nca_analysis()` on the rescaled constructs. NCA provides necessity effect sizes, permutation p-values, and bottleneck thresholds for the chosen target outcome level. Finally, the PLS-SEM and NCA results are merged into a tabular summary and a bubble plot that implements the cIPMA logic.

The package exports two user-facing functions: `cipma()` for theory-driven cIPMA on a bootstrapped PLS-SEM model and `get_rescaled_scores()`, which returns 0–100 rescaled latent scores from any `seminr` model given indicator minima and maxima. Two S3 methods complete the interface: `print.cipma()` displays the main numerical results, and `plot.cipma()` generates the cIPMA bubble map using `ggplot2`. Together, these components yield a single R object that stores all numerical and graphical outputs and can be saved, shared, and reused in fully reproducible workflows.

2.1. Requirements and assumptions

For cIPMA results to be valid and interpretable, several methodological requirements must be satisfied. Table 1 summarizes these requirements along with their rationale and verification methods. Researchers should verify these conditions before interpreting cIPMA results.

2.2. Composite scores and performance scaling

Let $\mathbf{x}_i = (x_{i1}, \dots, x_{ip})'$ denote the indicators for case i. Suppose construct j is measured by indicators indexed by M_j. In `seminr`, outer weights w_{jk} are estimated such that the composite score for case i and construct j is

$$\eta_{ij} = \sum_{k \in M_j} w_{jk} x_{ik}.$$

In matrix form, with data matrix $\mathbf{X}$ $(n \times p)$ and weight matrix $\mathbf{W}$ $(p \times m)$, the matrix of construct scores is

$$\boldsymbol{\eta} = \mathbf{X}\mathbf{W}.$$

`seminr` iteratively updates the weights and the structural path coefficients (by default using the path-weighting scheme) until convergence,

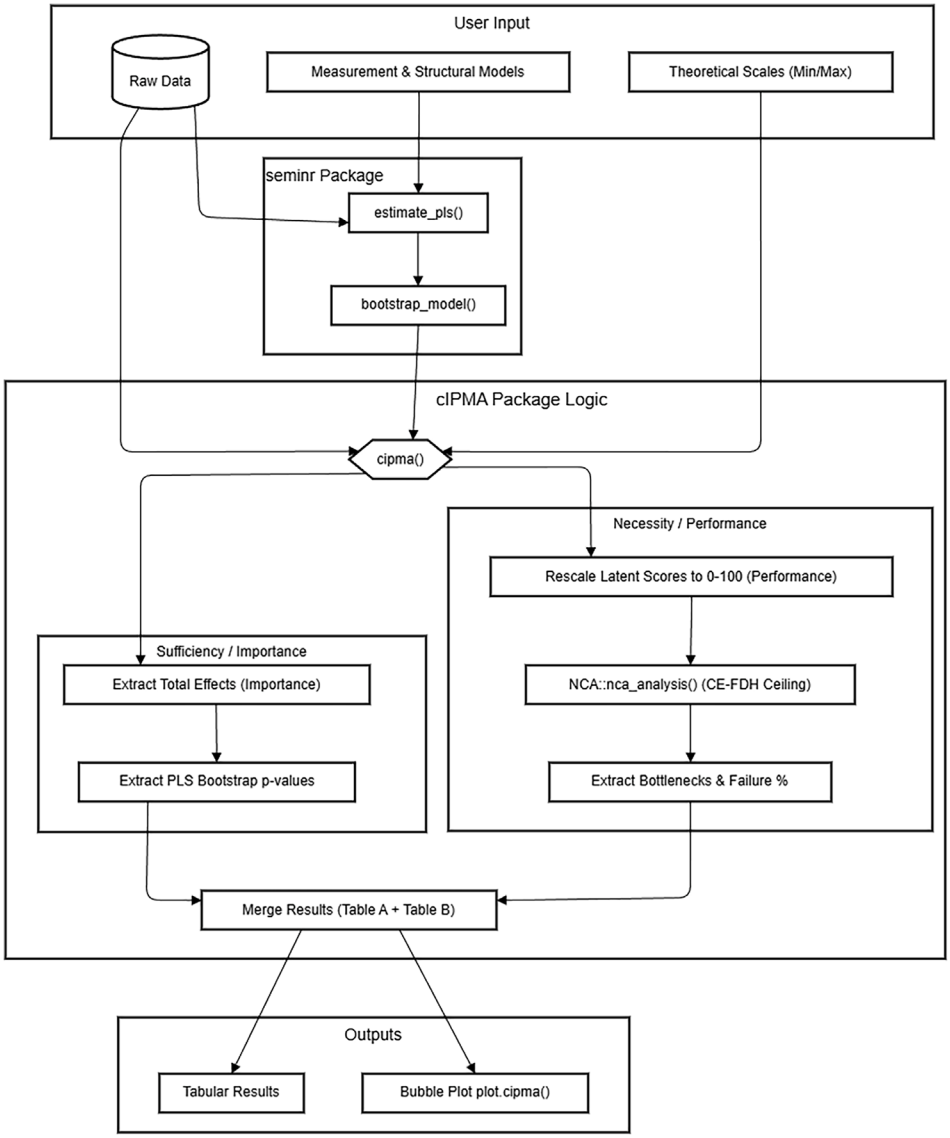

Fig. 1. Schematic workflow of the `cIPMA` package.

Table 1
Methodological requirements for valid PLS–SEM in cIPMA application.

Requirement	Rationale	How to verify	Consequence if violated
Positive outer weights	Negative weights invert the performance scale, making high scores appear as low performance	Check `seminr` output for outer weights; all should be positive	Misleading performance scores
Well-defined indicator scales	Performance rescaling requires known theoretical minima and maxima	Ensure all indicators have documented scale endpoints (e.g., 1–5 Likert)	Invalid 0–100 transformation
Theory-driven structural model	cIPMA importance is based on total effects from a specified model	Justify paths from substantive theory before estimation	Spurious importance estimates
Adequate sample size	Both PLS-SEM and NCA permutation tests require sufficient observations	Follow PLS-SEM guidelines (e.g., 10× rule); NCA recommends $n \geq 50$	Unstable estimates, inflated Type I/II errors
No multicollinearity	High VIF values distort path coefficients and total effects	Check VIF < 5 (conservative: < 3) in `seminr`	Unreliable importance rankings
Measurement model validity	Composites must adequately capture latent constructs	Assess reliability (Cronbach's α, composite reliability) and validity (AVE, HTMT)	Invalid construct scores

and returns the standardized construct scores, the outer weight matrix, and the path coefficients. For cIPMA, construct scores must be interpretable on a common 0–100 "performance" scale. The package reconstructs unstandardized outer weights by dividing each standardized weight $w_{\text{std},jk}$ by the empirical standard deviation s_k of indicator k,

$$\tilde{w}_{jk} = \frac{w_{\text{std},jk}}{s_k}, \quad s_k > 0,$$

and then normalizes these unstandardized weights,

$$v_{jk} = \frac{\tilde{w}_{jk}}{\sum_{\ell \in M_j} \tilde{w}_{j\ell}}.$$

The resulting weights v_{jk} are applied to the original (unscaled) indicators to obtain composite scores on the raw measurement scale. These composites are then transformed to a 0–100 range using the theoretical indicator bounds for each construct. If construct j is measured by Likert items with minimum a_j and maximum b_j, and η_{ij} is its composite for case i, the rescaled performance score is

$$\eta^*_{ij} = 100 \times \frac{\eta_{ij} - a_j}{b_j - a_j}.$$

This procedure yields performance scores that are comparable across constructs even when their indicators were originally defined on different scales.

The rescaling procedure described above assumes that all outer weights used in cIPMA are non-negative. When an outer weight is negative, the direction of the indicator–construct relationship is reversed and that indicator contributes in the opposite direction to the composite. In a performance context, this would effectively invert that indicator's contribution: higher raw scores would lower the rescaled performance score. If negative outer weights are detected, researchers should examine whether the indicator is reverse-coded or whether there is a substantive reason for the negative loading, and consider revising the measurement model. The `cIPMA` package issues a warning when negative outer weights are encountered.

2.3. NCA effect sizes and bottlenecks

For each predictor–outcome pair, `cIPMA` runs `NCA::nca_analysis()` with the CE-FDH ceiling technique on the rescaled composite scores. NCA characterizes necessity through the "empty space" in the upper-left region of an **X**–**Y** scatterplot: the ceiling line $y = f(x)$ separates the feasible area from a region in which observations are absent. The necessity effect size d is defined as the ratio of the ceiling zone area C to the total data scope S,

$$d = \frac{C}{S}, \qquad 0 \le d \le 1.$$

Statistical significance is assessed via permutation. The outcome values **Y** are permuted B times while the predictor **X** is held fixed; an effect size d_{perm} is recomputed for each permutation, and a p-value is obtained as

$$p = \frac{\#\{d_{\text{perm}} \ge d_{\text{obs}}\}}{B},$$

where d_{obs} is the observed effect size. This test guards against empty spaces produced by chance in unrelated variables [11,12]. For a user-specified target outcome level Y^*, the ceiling line is inverted to obtain the minimum predictor level $X^* = f^{-1}(Y^*)$. Cases with $X < X^*$ are counted as bottleneck failures, and their percentage is used as the third dimension (bubble size) in the cIPMA map.

In summary, `cipma()` automates the core procedure proposed by Hauff et al. [1]: it extracts total effects from a bootstrapped PLS-SEM model to quantify importance, computes mean rescaled composite scores to quantify performance, runs NCA to obtain necessity effect sizes and permutation-based p-values, and derives bottleneck thresholds and failure rates for a chosen target outcome level. The combined results are returned in a single object that can be printed, plotted, and reused in fully reproducible analyses.

2.4. Interpretation guidelines

cIPMA combines two complementary perspectives: sufficiency-based importance from PLS-SEM and necessity-based constraints from NCA. The importance dimension is based on PLS-SEM total effects and reflects how strongly, on average, a one-unit increase in a predictor is associated with a change in the outcome (including both direct and indirect paths). High importance, therefore, signals a promising lever for improving the expected outcome, but it does not imply that raising this predictor alone guarantees high outcomes for all cases. The necessity dimension instead captures minimum threshold levels that must be reached before high outcomes become feasible. A predictor with a substantive necessity effect size (d) acts as a bottleneck: if it falls below its required threshold, high outcome levels remain unattainable regardless of the status of other predictors.

The cIPMA framework constructs a two-dimensional map defined by *importance* (total effects, x-axis) and *performance* (average latent variable scores, y-axis). Necessity logic is superimposed onto this grid rather than defining the axes themselves: visual codes (e.g., color) distinguish necessary conditions, while *bubble size* encodes the bottleneck failure rate—specifically, the percentage of cases falling below the required necessary level. This overlay fundamentally shifts the interpretation of the standard four IPMA quadrants. Most notably, predictors located in traditionally lower-priority zones (e.g., low importance) must be re-prioritized for immediate intervention if they are identified as unsatisfied necessary conditions [1].

2.5. Practical usage notes

The `cIPMA` package is designed to keep model assessment and cIPMA reporting in one place. Because `cipma()` retains the bootstrapped PLS-SEM object, users can obtain standard diagnostics—such as Cronbach's α, composite reliability, AVE, HTMT, and path significance—directly via `result$PLS_Model` without re-estimating the model. This coupling of SEM diagnostics and cIPMA output supports transparent reporting and simplifies reproducible workflows.

The `plot.cipma()` method uses adaptive axis scaling rather than fixed 0–100 bounds. When performance scores cluster in a narrow range (for example, 60–80), the function automatically zooms in so that differences between constructs remain visible. The vertical and horizontal reference lines are placed at the mean importance and mean performance of the plotted constructs, yielding a relative importance–performance map. This helps users quickly see which constructs perform above or below the model average, even when absolute performance differences are small, and supports more nuanced managerial recommendations.

3. Illustrative examples

To illustrate the usage of cIPMA, consider a PLS–SEM study based on an extended technology acceptance model (TAM), used in previous empirical examples in the original cIPMA publications [13]. The researcher specifies reflective composites for constructs such as `Perceived_Usefulness`, `Compatibility`, `Ease_of_Use`, `Emotional_Value`, and `Adoption_Intention`, each measured on a 1–5 Likert scale, with the outcome `Technology_Use` measured on a 1–7 Likert scale. First, the user can define the measurement model using `seminr` syntax. Then, after estimating and bootstrapping the PLS-SEM model, the user can call `cipma()` to perform the combined analysis. Below is a complete R script demonstrating this process:

```
library(cIPMA)
#library(seminr)
#library(NCA)
#library(ggplot2)
#library(ggrepel)

# 1. Load Data
tam_data <- read.csv("tam.csv", check.names = FALSE)

# 2. Define seminr Model
tam_mm <- constructs(
  composite("Perceived_Usefulness", multi_items("PU_0", 1:3)),
  composite("Compatibility",        multi_items("CO_0", 1:3)),
  composite("Ease_of_Use",          multi_items("EOU_0", 1:3)),
  composite("Emotional_Value",      multi_items("EMV_0", 1:3)),
  composite("Adoption_Intention",   multi_items("AD_0", 1:3)),
  composite("Technology_Use",       single_item("USE_01"))
)

tam_sm <- relationships(
  paths(from = c("Compatibility", "Perceived_Usefulness",
  "Ease_of_Use", "Emotional_Value"),
  to = "Adoption_Intention"),
  paths(from = c("Adoption_Intention", "Compatibility",
  "Perceived_Usefulness", "Ease_of_Use", "Emotional_Value"),
  to = "Technology_Use")
)

# 3. Estimate \& Bootstrap (REQUIRED for cIPMA CIs)
# Note: cIPMA requires bootstrapped model
tam_pls <- estimate_pls(tam_data, tam_mm, tam_sm)
boot_tam <- bootstrap_model(tam_pls, nboot = 1000)

# 4. Define Scales for Rescaling (Theoretical Min/Max)
my_scales <- list(
  PU_01 = c(1, 5), PU_02 = c(1, 5), PU_03 = c(1, 5),
  CO_01 = c(1, 5), CO_02 = c(1, 5), CO_03 = c(1, 5),
  EOU_01 = c(1, 5), EOU_02 = c(1, 5), EOU_03 = c(1, 5),
  EMV_01 = c(1, 5), EMV_02 = c(1, 5), EMV_03 = c(1, 5),
  AD_01 = c(1, 5), AD_02 = c(1, 5), AD_03 = c(1, 5),
  USE_01 = c(1, 7)
)

# 5. Run cIPMA
result <- cipma(
  model = boot_tam,
  target_construct = "Technology_Use",
  data = tam_data,
  scales = my_scales,
  target_level = 85,
  pls_model = tam_pls
)

# 6. Outputs
print(result)
plot(result)
plot1 <- plot(result)

# 7. Save as PNG with specific dimensions
ggsave(
  filename = "cipma_plot.png",
  plot = plot1,
  width = 700,
  height = 550,
  units = "px",
  dpi = 100,
  bg = "white"
)
```

3.1. Tabular cIPMA output and reporting

The `cipma()` function returns an object of class `cipma` containing all numerical results. The two main summary tables are:

- `result$cIPMA_results` (*Table A*): PLS-SEM importance, performance, significance, NCA effect size d, necessity flag, and failure percentages for each predictor.
- `result$Bottleneck_summary` (*Table B*): required predictor levels at the target outcome level, with case counts failing to meet each threshold.
- `result$Bottleneck_full`: complete bottleneck table (0–100% in 5% increments)
- `result$NCA_summary`: NCA analysis artifact from the `NCA` package.
- `result$Data_Rescaled`: construct scores rescaled to 0–100
- `result$PLS_Model`: the underlying `seminr` model for additional diagnostics.

These components can be inspected directly in R:

```
result <- cipma(\ldots)
# Table A: combined PLS-SEM and NCA results
print(result$cIPMA_results)
# Table B: bottleneck summary
print(result$Bottleneck_summary)

=== Combined Importance-Performance Map Analysis (cIPMA) ===
Target Construct: Technology_Use
```

```
Desired Outcome Level: 85

--- Table A: PLS-SEM \& NCA Statistics ---
             Construct Importance Performance  PLS_p_value     NCA_d NCA_p_value Is_Necessary
1   Adoption_Intention 0.43705233    72.04067 1.344544e-06 0.2941647      0.0002          TRUE
2        Compatibility 0.12699788    61.55690 3.026167e-01 0.2110126      0.0001          TRUE
3          Ease_of_Use 0.04863022    75.64040 5.532847e-01 0.2350482      0.0150          TRUE
4      Emotional_Value 0.36169649    70.17135 7.960791e-06 0.3310017      0.0001          TRUE
5 Perceived_Usefulness 0.14921041    64.24785 1.729285e-01 0.2429061      0.0009          TRUE

--- Table B: Bottleneck Summary ---
             Construct Required_Level Fail_Percentage Fail_Count
1        Compatibility         33.696         8.620690         15
2 Perceived_Usefulness         66.212        47.126437         82
3          Ease_of_Use         66.904        28.735632         50
4      Emotional_Value         49.650         5.747126         10
5   Adoption_Intention         75.000        39.080460         68
```

For publication, researchers may wish to merge selected columns from Table A and Table B into a single summary table. Table 2 shows an example of such a combined report for the target construct `Technology_Use` with a target level of 85 on the 0–100 scale. The columns make explicit the sufficiency-based importance measure (PLS-SEM total effects τ_{jT}), the performance metric (mean rescaled construct score $\bar{\eta}^*$), the necessity effect size d, and the bottleneck information (required level X^* and percentage of failing cases).

In this theory-driven specification, all five antecedents are modeled as direct predictors of `Technology_Use`. Fig. 2 shows that `Adoption_Intention` has the highest importance (total effect $\tau_{jT} \approx 0.44$) and relatively high performance, followed by `Emotional_Value`. `Perceived_Usefulness` and `Compatibility` exhibit lower performance and smaller total effects, whereas `Ease_of_Use` shows high performance but low importance. Table 2 combines these PLS-SEM results with NCA information. All constructs satisfy the necessity criterion ($d \geq 0.10$ and $p < 0.05$), but they differ markedly in bottleneck severity: almost half of the cases (47%) fail to reach the required level of `Perceived_Usefulness`, while only about 6% fail the `Emotional_Value` bottleneck. From a cIPMA perspective, `Adoption_Intention` and `Perceived_Usefulness` emerge as high-priority levers because they combine relatively high importance with substantial bottleneck failure rates.

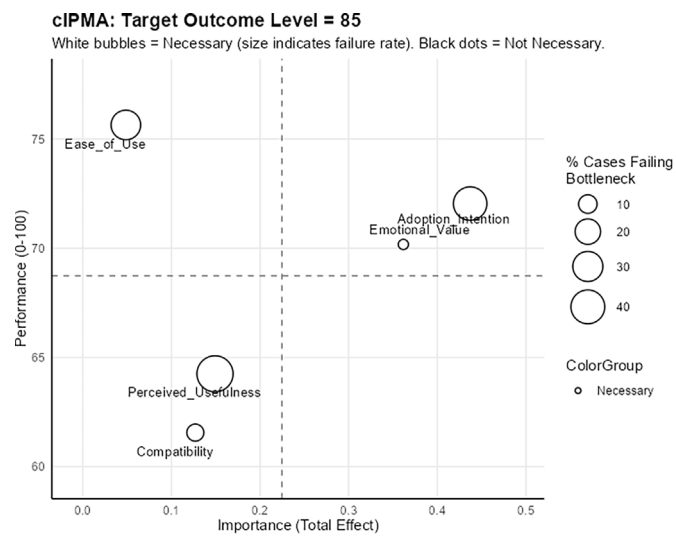

Fig. 2. Results of the cIPMA analysis showing the performance and importance of constructs.

3.1.1. Assumption checking and diagnostics

Before finalizing the interpretation, it is crucial to verify that the underlying PLS-SEM model meets the methodological requirements for cIPMA. The `check_assumptions()` function automates this process by validating outer weights, reliability, discriminant validity, multicollinearity, and sample size against established thresholds.

```
# Check assumptions and model diagnostics
check_assumptions(result)
```

The output below highlights a specific issue in the illustrative model:

```
=== cIPMA Assumption Checks ===

1. POSITIVE OUTER WEIGHTS
   [PASS] All outer weights are positive.

2. RELIABILITY (alpha >= 0.70, rhoC >= 0.70, AVE >= 0.50)
                      alpha  rhoC   AVE  rhoA
Compatibility         0.858 0.914 0.779 0.859
Perceived_Usefulness  0.723 0.842 0.642 0.753
Ease_of_Use           0.783 0.873 0.697 0.783
Emotional_Value       0.914 0.946 0.853 0.917
Adoption_Intention    0.938 0.960 0.889 0.939
Technology_Use        1.000 1.000 1.000 1.000

Alpha, rhoC, and rhoA should exceed 0.7 while AVE should exceed 0.5
   [PASS] All reliability thresholds met.
```

Table 2
Combined importance-performance map analysis (cIPMA) results for `Technology_Use` (Target Level = 85).

Predictor construct	Importance	Performance	PLS p-value	NCA d	NCA p-value	Necessary?	Required Level	Fail percentage
	(τ_{jT})	($\bar{\eta}^*$)	(Sufficiency)	(Effect size)	(Necessity)	($d \geq 0.10, p < 0.05$)	($X^* \geq \hat{f}_X(85)$)	(% cases < X^*)
Adoption Intention	0.437	72.04	$< 0.001^{***}$	0.294	$< 0.001^{***}$	**Yes**	75.00	39.08%
Emotional Value	0.362	70.17	$< 0.001^{***}$	0.331	$< 0.001^{***}$	**Yes**	49.65	5.75%
Perceived Usefulness	0.149	64.25	0.173	0.243	0.001^{***}	**Yes**	66.21	47.13%
Compatibility	0.127	61.56	0.303	0.211	$< 0.001^{***}$	**Yes**	33.70	8.62%
Ease of Use	0.049	75.64	0.553	0.235	0.015^{*}	**Yes**	66.90	28.74%

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. p-values are based on bootstrap (PLS) and permutation (NCA) tests.

```
3. DISCRIMINANT VALIDITY (HTMT < 0.9 )
                     Compatibility Perceived_Usefulness Ease_of_Use Emotional_Value Adoption_Intention Technology_Use
Compatibility                    .                    .           .               .                  .              .
Perceived_Usefulness         0.961                    .           .               .                  .              .
Ease_of_Use                  0.526                0.594           .               .                  .              .
Emotional_Value              0.675                0.539       0.549               .                  .              .
Adoption_Intention           0.631                0.637       0.524           0.739                  .              .
Technology_Use               0.517                0.489       0.375           0.551              0.642              .
   [FAIL] HTMT exceeded for:
          Perceived_Usefulness <-> Compatibility

4. MULTICOLLINEARITY (VIF < 5 )
   Adoption_Intention:
    Compatibility = 3.05, Perceived_Usefulness = 2.54, Ease_of_Use = 1.41, Emotional_Value = 1.71
   Technology_Use:
    Adoption_Intention = 2.17, Compatibility = 3.05, Perceived_Usefulness = 2.65, Ease_of_Use = 1.42, Emotional_Value = 2.28
   [PASS] No multicollinearity issues detected.

5. SAMPLE SIZE
   N = 174 (recommended >= 60 )
   [PASS] Sample size is adequate.
```

In this example, the diagnostic check reveals a discriminant validity failure (HTMT = 0.961) between `Perceived_Usefulness` and `Compatibility`. While the cIPMA plot correctly displays the calculated importance and performance scores, this warning indicates that the two constructs are empirically indistinguishable in this dataset. Consequently, the "importance" estimates (total effects) for these variables may be confounded.

4. Impact

The `cIPMA` package contributes to strategic management and behavioral research by operationalizing the combined sufficiency–necessity logic of cIPMA in a single, scriptable environment. It removes dependence on commercial software and manual file transfers, and it reduces specification and transcription errors by automating indicator rescaling, performance score construction, NCA configuration (including CE-FDH ceilings and permutation testing), and bottleneck extraction [1].

The package also introduces diagnostic tools to promote methodological rigor. The `check_assumptions()` function automates verification of critical criteria for interpretable cIPMA results: outer weight signs, construct reliability (α, ρ_C, AVE, ρ_A), discriminant validity via HTMT ratios, multicollinearity via VIF, and sample size adequacy. These checks help users detect misspecification risks before interpreting importance, performance, and bottleneck findings. The documentation further summarizes key requirements (positive outer weights, well-defined indicator scales, adequate sample size, and theory-driven model specification), consistent with guidance in the cIPMA literature [1].

Because both IPMA and NCA are fundamentally visual decision-support tools, reproducible visualization is central to their applied value. The `plot.cipma()` function produces standardized, publication-ready cIPMA maps that can be regenerated exactly from code and version-controlled alongside analysis scripts.

In addition, the package provides experimental support for structure learning via `discovery_cipma()`, interfacing with `bnlearn` [9,10]. This extension supports score-based, constraint-based, and hybrid algorithms and allows users to incorporate domain knowledge via whitelist and blacklist constraints. Recent work on combining PLS-SEM with machine learning illustrates how discovery-oriented methods can complement theory-driven modeling [8]. However, integrating probabilistic structure learning based on conditional independence with NCA's boundary-based necessity logic raises unresolved methodological questions, motivating further research rather than definitive software claims. Related developments on probabilistic notions of necessity and sufficiency may offer alternative foundations for future bottleneck modeling [14].

5. Conclusions

This paper introduces `cIPMA`, an R package that unifies sufficiency-based PLS-SEM importance with necessity-based NCA constraints in a single analytical workflow. While cIPMA has gained traction as a managerial decision-support tool, its implementation previously required coordinating multiple software environments. By providing automated rescaling, bottleneck computation, assumption checking, and publication-ready visualization, `cIPMA` lowers the barrier to methodologically sound and reproducible application [1].

The implementation follows established cIPMA interpretation rules [1,6]. Importance is computed from PLS-SEM total effects and therefore captures both direct and indirect (mediated) pathways. Necessity is assessed bivariately between each predictor and the focal outcome using NCA, yielding ceiling-based necessity effect sizes and bottleneck thresholds that are not conditional on other predictors [3]. Consequently, the current framework treats necessity as a direct predictor–outcome constraint rather than propagating bottlenecks through mediated chains. Formal rules for "cascaded necessity" (i.e., how necessity constraints should propagate along mediated or discovered causal chains) remain an open methodological question.

The experimental structure-learning component highlights a promising avenue for future research: DAG-adaptive cIPMA. A key unresolved issue is whether necessity constraints should propagate sequentially through discovered causal chains (requiring bottleneck decomposition at each step) or whether the current approach—direct bivariate NCA combined with total-effect importance—provides an adequate practical approximation in mediated structures. While `cIPMA` focuses on the established theory-driven workflow, its architecture provides a foundation for testing these extensions as they mature in the methodological literature, and for developing formal bottleneck propagation rules as part of future methodological work [8,14].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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